جامعة التقنية والعلوم التطبيقية
University of Technology and Applied Sciences
نزوى
Nizwa

PREPARATORY STUDIES CENTRE

FINAL- EXAMINATION Set A

SEMESTER FALL, ACADEMIC YEAR 2023 –2024

MATH1202 - CALCULUS I

Day and Date: Thursday/ 04 -01-2024 Time: 9:00am: 11:00am

Max. Marks	50	Time Allocation	2 hours
Student ID		Student Name	
Section		Level	Diploma

Instructions to the Students

1. This exam paper consists of 10 pages including the front page. If any page is missing, inform the invigilator immediately.
2. Answer the questions in the space provided by using **a blue or black** pen only. You will obtain azero (0) if you use a pencil for all questions.
3. Use of mobile phones and any electronic gadgets inside the examination hall are not allowed.
4. Scientific calculators (not programmable) are allowed.
5. Copying, Cheating, and any kind of malpractice on the examination are strictly prohibited. Actionwill be taken as per the "Article 68, Bylaws of UTAS Academic Regulations".

Question Type	Allotted Marks	Obtained Marks
PART A -Multiple Choice Questions	$10 \times 1 = 10$	
PART B - Short Answer Questions	$5 \times 3 = 15$	
PART C - Long Answer Questions	$5 \times 5 = 25$	
Total Marks	50	

First Marker	Second Marker

PART A: MULTPLE CHOICE QUESTIONS

(Choose the correct answer. Each question carries 1 mark)

(10 x 1 = 10 marks)

1. $\lim_{x\to 3} \frac{x^2-3x-4}{x^2-7} =$ _____

(a) 2 (b) 4 (c) 1 (d) -2

2. $\lim_{x\to 7}(\ln e^{2x}) =$ _____

(a) 14 (b) e (c) 2 (d) 7

3. The horizontal asymptote for the function $f(x) = \frac{3x^2+2x}{4x^3-5x+7}$ is y = _____

(a) $\frac{3}{4}$ (b) 0 (c) ∞ (d) $\frac{2}{3}$

4. If the position of the particle is $s(t) = 3t^3 + t - 15$ meters, then the instantaneous velocity of the particle at $t = 3$ seconds is _____ m/sec.

(a) 82 (b) 28 (c) 69 (d) 81

5. If $f(x) = 7\tan^{-1} x$, then $f'(x) =$ _____

(a) $\frac{7}{1-x^2}$ (b) $\frac{-7}{1+x^2}$ (c) $\frac{7}{1+x^2}$ (d) $\frac{7}{x^2-1}$

6. If $x = t^4$ and $y = t$, then $\frac{dy}{dx} =$ _____

(a) $\frac{-1}{4t^3}$ (b) $\frac{1}{4t^3}$ (c) $4t^3$ (d) $\frac{-1}{t^3}$

7. The volume of a cube increases at a rate of $12m^3/s$, when the side of the cube is $2m$. The rate at which the side increases is _____

(a) $-2\ m/s$ (b) $-1\ m/s$ (c) $4\ m/s$ (d) $1\ m/s$

8. If $f(x) = 2x^3$, the value of $'c'$ by using mean value theorem in the interval [0, 2] is _____

(a) -1.1547 (b) 1.1547 (c) 0.4330 (d) 1.3333

9. $\int 3sec^2x\ dx =$ _____

(a) $3\tan x + c$ (b) $3\cot x + c$ (c) $3secx \tan x + c$ (d) $3\,csc^2\,x + c$

10. Using integration by parts, $\int xe^x dx =$ _____

(a) $e^x - x + c$ (b) $xe^x + e^x + c$ (c) $xe^x - e^x + c$ (d) $xe^x + c$

Part-B: SHORT ANSWER QUESTIONS
(Answer all questions. Each question carries 3 marks)

(5 x 3 = 15 marks)

11. Find the equation of the tangent line to the graph of $xy^2 + \sin(\pi y) - 2x^2 = 2$

at the point $(1, 2)$.

[OR]

Find the derivative of $y = (3x^4 + 2)^{\tan x}$ by using logarithmic differentiation.

12. Anwar needs to construct a rectangular pen around an area of $2500ft^2$.
    a) What are the dimensions of the rectangular pen to minimize the amount of material needed?
    b) Find the possible perimeter of a pen.

13. Evaluate: $\lim_{x \to 0} \frac{12\, e^x - 12x - 12}{x^2}$ by using L' Hospital's Rule.

14. A rocket is fired vertically upward from the ground. The distance in meter that the rocket travels from the ground after t seconds is given by $s(t) = 2t^3 + 7t^2 - 11t + 213$ meters.

    a) Determine the velocity of the rocket at $t = 5$ sec.

    b) Determine the acceleration of the rocket at $t = 4$ sec.

15. Find the value of k that makes the function $f(x) = \begin{cases} \frac{\sqrt{x+4}-1}{x+3}, & x \neq -3 \\ k, & x = -3 \end{cases}$ is continuous at $x = -3$

PART C: LONG ANSWER QUESTIONS
(Answer the question. Each question carries 5 marks)

(5 x 5 = 25 marks)

16. Verify that the function $f(x) = x^3 - 6x^2 + 11x - 6$ defined over the interval $[1, 3]$ satisfies the conditions of Rolle's Theorem. Find all points guaranteed by Rolle's Theorem.

[OR]

A ball is dropped from a height of $224\ feet$. Its position at time t seconds is $s(t) = -16t^2 + 224$. Find the time t when the instantaneous velocity of the ball equals its average velocity.

17. Consider $f(x) = 2x^3 - 6x^2 + 12$. Determine the following.

a) The intervals where the function is increasing or decreasing.
b) The point(s) of local maxima or minima.
c) The intervals where the function is concave up or concave down.
d) The inflection point(s).

18.(a) Evaluate the integral: $\int(-\csc x \cot x + \frac{9}{\sqrt{1-x^2}})\, dx$ [2 marks]

18.(b) Solve the initial value problem: $f'(x) = x^3 - 8x^2 + 1,\ \ f(1) = 1$ [3 marks]

19. Find $\frac{dy}{dx}$ for the following functions.

(a) $y = \ln(9x^3 + 29) + (4 + cotx)^5$, by using chain rule. [3 marks]

(b) $y = \frac{x^2 + 4}{x^2 - 6}$ [2 marks]

20. (a) Using substitution method, evaluate the integral: $\int x(1+x)^{98}dx$ [3 marks]

20. (b) Evaluate: $\int_{-3}^{4}(t+2)(t-3)dt$ [2 marks]